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Software Test Plan
e-Sikho

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Table of Contents

Document History 4

1. Introduction 5
   - 1.1 Purpose 5
   - 1.2 Scope 5
   - 1.3 Out of Scope 5
2. Test Objectives 5
3. Test Strategy 6
   - 3.1 Testing Types 6
   - 3.2 Test Environment 6
   - 3.3 Test Data Strategy 6
   - 3.4 Defect Strategy 6
4. Test Cases by Module 7
   - M1 - Users Authentication and Profile 7
   - M2 - Courses and Enrollment 7
   - M3 - Assessment Hosting Applications 8
   - M4 - Audience Eligibility 9
   - M5 - Exams and Grading 10
   - M6 - Contests 10
   - M7 - Video Resources and Live Classes 11
   - M8 - Administration and Regression 12
5. Test Coverage Summary 13
6. Pass / Fail Criteria 13
   - 6.1 Pass Criteria 13
   - 6.2 Fail Criteria 13
7. Test Schedule 14
8. Tools & Dependencies 14
9. Conclusion 15

Document History

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1. Introduction

This Test Plan defines the manual testing strategy, scope, objectives, cases and schedule for e-Sikho, a web learning platform built with Next.js, React and MongoDB. Students access courses, exams and contests; instructors author learning content; administrators control privileged workflows.

1.1 Purpose

Provide repeatable manual procedures and expected results for the project team. The 69 planned cases verify functional behavior and authorization. Their initial execution status is Not Run; this document does not assert manual test success.

1.2 Scope

The test plan covers eight application modules:

- M1 - Users Authentication and Profile
- M2 - Courses and Enrollment
- M3 - Assessment Hosting Applications
- M4 - Audience Eligibility
- M5 - Exams and Grading
- M6 - Contests
- M7 - Video Resources and Live Classes
- M8 - Administration and Regression

1.3 Out of Scope

- Load/stress testing, penetration testing and institutional identity verification.
- Real payment settlement/refund/payout transfers while provider integration is unavailable.
- Full DRM certification or protection for external public media; unsafe code sandbox redesign.

2. Test Objectives

- Verify protected pages, ownership, course entitlement and role restrictions.
- Validate publishing, approval, audience matching, scoring and time-window rules.
- Confirm actual data persists and historical samples are not recreated.
- Record browser errors, provider blockers and reproducible defects with evidence.

3. Test Strategy

3.1 Testing Types

Test Type	Tool / Framework	Description
Functional / usability	Manual browser testing	Execute positive/negative forms, navigation, scoring and responsive checks.
Access / integration	Browser profiles + DevTools	Compare roles, entitlements and cross-module persistence; inspect protected responses.
Provider end-to-end	Browser + provider console	Verify actual OTP/OAuth/media flows when configured; otherwise record Blocked.

3.2 Test Environment

- OS: Windows 10/11; desktop Chrome and one mobile browser.
- Application: Next.js 14.2.35, React 18; Node.js 24.19.0 used in the prior verification.
- Database: isolated staging MongoDB; never restore samples to the real database.
- Local start: npm run dev; use the printed app URL, normally localhost:3000.
- Record branch/commit, uncommitted changes, browser and Asia/Dhaka timestamps.

3.3 Test Data Strategy

Use separate sessions: SA superadmin; AD ordinary admin; I1 course owner; I2 unassigned instructor. Students: S1/S2 School Science; S3 College Science; S4 University CSE; S5 University EEE; S6 unset profile. Only S1 is selected for the targeted school exam. Education details are self-reported.

C1 belongs to I1; S1 enrolled, S2 unenrolled. S4 has C2 access only. E1 has three 2-mark MCQs, pass mark 2, deduction 0.25 and two attempts. GE1 targets School + Science + S1; GC1 targets University + CSE. Use separate exams or wait through the 24-hour cooldown.

3.4 Defect Strategy

Record case ID, actual result, Pass/Fail/Blocked/N/A, tester and evidence in the results sheet. Defects include steps, expected/actual behavior, role, environment, severity and redacted screenshot/log. Retest fixes; never copy automated passes into manual results.

4. Test Cases by Module

TC-Mxx identifies a manual case. Type Manual means functional/usability; Access means a manual authorization/security check. All cases begin Not Run. Check both exams and contests for M4 rules; no banner or direct-link bypass is acceptable.

M1 - Users Authentication and Profile

Registration, login, logout, role boundaries and student profile persistence.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M01	Successful login	S1 signs in; reload /student.	Correct student session and identity persist.	Manual
TC-M02	Invalid credentials	Try wrong password, unknown account and incomplete OTP.	No authenticated session or replacement account is created.	Access
TC-M03	Signup and OTP	Register in staging with wrong, then valid OTP; retry registration.	Only valid verification creates one account; duplicates rejected.	Manual
TC-M04	Anonymous route access	Signed out: open student, instructor, admin and payment URLs.	Sign-in required; no protected data returned.	Access
TC-M05	Role boundaries	S1 opens author/admin routes; I1 and AD open /admin/assessments.	Students cannot author; only superadmin reviews global hosting.	Access
TC-M06	Logout and session access	Sign out; use Back, reload and retry a protected action.	Protected actions require a fresh authenticated session.	Access
TC-M07	Blocked account	SA blocks a staging account; refresh its active session.	Blocked account cannot continue protected use.	Access
TC-M08	Google sign in	Use configured Google login; verify role; cancel a second sign-in.	Existing role/status retained; cancellation has clear feedback.	Manual
TC-M09	Student profile values	Save education and department in /student/settings; reload.	Real values persist; blank profile gets no invented background.	Manual

M2 - Courses and Enrollment

Course authoring, visibility, preview/private access, ownership and revision entitlement.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M10	Course authoring	I1 creates C1, module and lesson; reopens editor.	Content persists under I1; no sample content is added.	Manual
TC-M11	Draft visibility	While C1 is draft, check public catalog, URL and course API.	Draft content withheld; no public enrollment.	Access

M2 - Courses and Enrollment continued

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M12	Published catalog	Publish C1; find it in /courses and open details.	Correct published title, details and links appear.	Manual
TC-M13	Preview and private lessons	Without enrollment, try preview/private lessons; inspect metadata.	Only configured previews open; no private media URL in metadata.	Access
TC-M14	Free enrollment	S1 enrolls in free published C1; reload and repeat.	Access granted once; no duplicate enrolled-course entry.	Manual
TC-M15	Curriculum ownership	I2 attempts to edit/delete I1's modules or lessons.	Mutation denied; original content unchanged.	Access
TC-M16	Cross course access	S4 has C2 access only; try C1 private media/resources.	C2 entitlement never unlocks C1.	Access
TC-M17	Lesson completion	S1 completes a lesson through its supported control; reload.	Real completion persists for that student/course.	Manual
TC-M18	Expiry and revision access	Check expired C1 access, revision request, approval and expiry.	Request alone grants nothing; only active approved revision restores access.	Manual

M3 - Assessment Hosting Applications

Separate exam/contest permissions, superadmin review and instructor assignment.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M19	Course and scope selectors	I1 opens both builders before global approval.	Owned/assigned drafts selectable; global approval required.	Manual
TC-M20	No assigned courses	I2 opens builders with no owned/assigned courses.	Clear empty state; no silent global permission or selection.	Manual
TC-M21	Hosting application	Apply for global exams with a short, then 20–2000 character reason.	Short reason rejected; valid request PENDING; hosting locked.	Manual
TC-M22	Duplicate applications	Retry the same pending application; repeat after approval.	No duplicate pending/approved request for that type.	Manual
TC-M23	Exam approval	SA approves EXAM; I1 reloads exam and contest builders.	Global exam unlocks; global contest remains locked.	Access

M3 - Assessment Hosting Applications continued

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M24	Contest rejection	SA rejects CONTEST without notes, then with 10+ characters.	Insufficient notes rejected; feedback saved; contest stays locked.	Manual
TC-M25	Resubmission and contest approval	Resubmit rejected CONTEST; SA approves; reload builder.	Pending until reviewed; contest then unlocks independently.	Manual
TC-M26	Instructor assignment	SA assigns C1 assessment access to approved instructor I2.	I2 can author assessments; ownership/curriculum rights unchanged.	Manual
TC-M27	Invalid assignment and review access	Try assignment to student/unknown email; I2 tries approval.	Invalid assignment and instructor self-approval denied.	Access

M4 - Audience Eligibility

Education level, department and selected-student rules across discovery and participation.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M28	Specific student audience	GE1: School + Science + S1; test S1, S2, S3 on all surfaces.	Only S1 qualifies; S2 and S3 cannot view or participate.	Access
TC-M29	Department audience	GC1: University + CSE; test S4 and S5.	S4 qualifies; EEE student S5 has no access or banner.	Access
TC-M30	Combined filters	Choose School OR College plus Science; test S1, S3, S4.	S1/S3 qualify; filters combine AND across groups, OR within.	Access
TC-M31	Unrestricted registered audience	Publish with blank filters; test S6 and signed-out browser.	Registered S6 qualifies; anonymous participation denied.	Access
TC-M32	Department normalization	S4 uses spaced lowercase cse, then expanded CSE department name.	Recognized equivalent department labels match consistently.	Manual
TC-M33	Eligibility after profile change	Change S1 to Arts; refresh GE1 and retry old URL/submission.	New reads/submissions denied; stale client state grants nothing.	Access
TC-M34	Student selection	Search/select/remove a real student; also search unmatched text.	Only actual students selectable; no fabricated result; selection persists.	Manual

M5 - Exams and Grading

Authoring validation, entitlement, scoring, confidentiality, retakes and visible timer.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M35	Course exam authoring	I1 creates E1 for C1 with the documented scoring values.	Blank initial question; correct course, marks, duration and attempts saved.	Manual
TC-M36	Invalid exam inputs	Try empty fields, invalid options/marks, pass mark > total and invalid limits.	Invalid exams rejected with feedback; no malformed record saved.	Manual
TC-M37	Draft and enrolled access	Create for draft C1; publish; compare enrolled S1 and outsider S2.	Students require publication and current course entitlement.	Access
TC-M38	Negative marking calculation	E1: one correct 2-mark answer, one wrong, one skipped; submit.	1.75/6, 29%, Fail at pass mark 2; counts 1/1/1; skip costs 0.	Manual
TC-M39	Scoring boundaries	Separate exams: disable deduction; then submit all wrong with deduction.	Same pattern without deduction scores 2/6; all-wrong score floors at 0.	Manual
TC-M40	Retakes and cooldown	Submit E1; retry now and after 24 hours; exceed two attempts.	24-hour cooldown and two-attempt cap enforced; history persists.	Manual
TC-M41	Multiple question types	Test known True/False and short answers; benign CODE only in isolation.	Question types and scoring match key; unsafe judge remains a blocker.	Manual
TC-M42	Answer confidentiality	Before submission, inspect student responses and any preview control.	Correct answers, accepted answers and hidden test cases withheld.	Access
TC-M43	Visible exam timer	Let a short exam timer expire; record behavior and reload.	Advertised UI timeout behaves correctly; server timing is not certified.	Manual

M6 - Contests

Publication, valid time windows, submissions, solutions and banner status.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M44	Contest publication	Publish a permitted MCQ contest with real dates/questions.	Correct scope, audience, duration and unique URL saved.	Manual
TC-M45	Invalid contest inputs	Try reversed/ended window, excessive duration and empty questions.	Invalid contest rejected; useful validation shown.	Manual

M6 - Contests continued

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M46	Contest time window	Check before start, during window and after end; submit outside window.	Status follows real time; server rejects out-of-window submissions.	Access
TC-M47	Submission and results	Submit known answers live; reopen history/results and ranking.	Real student/contest result matches key; record duplicate-submit behavior.	Manual
TC-M48	Solutions and ended visibility	Inspect live responses; revisit home/catalog after contest ends.	Live solutions hidden; ended contest leaves home, remains eligible catalog.	Access

M7 - Video Resources and Live Classes

Provider settings, real playback, token protection, resources and live-session access.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M49	Provider settings	Inspect Bunny library 792504 Security settings without exposing keys.	Embed tokens/Basic DRM ON; MP4 fallback/originals/Early Play OFF.	Access
TC-M50	Real video upload	I1 uploads a small owned video; check encoding; reload lesson.	Real provider video and stable reference saved; errors explicit.	Manual
TC-M51	Authorized playback	Enrolled S1 plays, pauses, seeks and uses fullscreen; reload.	Authorized playback works without unrelated fallback media.	Manual
TC-M52	Private playback access	S2, anonymous and C2-only student try private C1 lesson.	App issues no playable private media; bearer URLs may last until expiry.	Access
TC-M53	Token integrity and expiry	Tamper with one token character; separately retry after real expiry.	Tampered/expired URLs cannot begin fresh playback; app can renew access.	Access
TC-M54	Protection and browser coverage	Check protected playback in Chrome and a mobile browser.	Verify provider evidence; UI controls alone prove no DRM.	Access
TC-M55	Private resources	S1/S2/anonymous try private C1 resource and a public preview.	Private app access denied to outsiders; no invented PDF fallback.	Access
TC-M56	Live session access	S1 and outsider S2 open a real C1 live session/recording.	Only entitled user receives private meeting/recording details.	Access

M7 - Video Resources and Live Classes continued

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M57	Live questions and management	S1 asks live question; I1 answers/pins; unrelated I2 retries.	Real question persists; only owner/staff can manage it.	Manual

M8 - Administration and Regression

Safe checkout state, administrative access, sample cleanup and browser usability.

TC-ID	Scenario	Input / Action	Expected Result	Type
TC-M58	Unavailable paid checkout	Open paid checkout and any old success/result link.	No fake success, PIN collection or enrollment granted.	Access
TC-M59	Administrative pages	Check admin refund/payout/ moderation/audit pages by role.	Nonadmins denied; allowed views show actual data/empty states.	Access
TC-M60	Instructor role application	Student applies; authorized reviewer approves instructor role.	No pending privileges; approval preserves user identity.	Manual
TC-M61	Empty content states	Browse home/catalog/dashboard on empty staging data.	Empty states; no fake content, stats or notifications.	Manual
TC-M62	Sample review and preserved accounts	Run read-only sample review; verify owner/admin access.	Zero unprotected matches; owner/admin kept; no deletions.	Access
TC-M63	No automatic reseeding	Compare content counts before/after read-only page visits.	Browsing never reseeds sample users/courses/as sessments.	Manual
TC-M64	Deleted sample links	Search removed sample titles and open their old URLs.	Old links fail safely; no sample recreation.	Manual
TC-M65	Responsive layout	Repeat major forms/pages at mobile 360px and desktop widths.	Readable layout; usable controls; no clipped submit button.	Manual
TC-M66	Language switching	Switch Bangla/English; reload; inspect values and errors.	Language consistent; untranslated labels logged.	Manual
TC-M67	Keyboard usability	Use keyboard through forms and modal controls.	Visible focus; named controls; workflow can be completed/exited.	Manual
TC-M68	Missing content and offline state	Open nonexistent URLs; simulate offline request; restore and retry.	Clear safe errors; no fake success/endless loading; recovery works.	Manual
TC-M69	Console and duplicate actions	Inspect Console/Network; repeat clicks, Back and reload.	No uncaught app errors or unintended duplicate records.	Manual

5. Test Coverage Summary

Module	Manual Tests	Access Tests	Total	Status
M1	4	5	9	Not Run
M2	5	4	9	Not Run
M3	7	2	9	Not Run
M4	2	5	7	Not Run
M5	7	2	9	Not Run
M6	3	2	5	Not Run
M7	3	6	9	Not Run
M8	9	3	12	Not Run
TOTAL	40	29	69	Not Run

These counts describe planned cases, not measured code coverage. Provider-dependent cases are Blocked until their setup exists; prior automated results are separate evidence.

6. Pass / Fail Criteria

6.1 Pass Criteria

- Actual behavior matches every expected check in a case.
- No unauthorized account can read private data or mutate another owner's records.
- Saved data, scores and eligibility remain consistent after reload.
- Record sanitized evidence; the run passes only with no unresolved failures or required blocked cases.

6.2 Fail Criteria

- Unauthorized access, leaked answers/media, forged enrollment or incorrect scoring.
- Valid data rejected or invalid data saved contrary to the expected behavior.
- Uncaught application errors, unintended duplicates or sample reseeding.
- Missing setup is Blocked, not Pass. Document N/A reasons and all remaining gaps.

7. Test Schedule

Phase	Activity	Deliverable
Phase 1	Prepare staging, accounts, content and build reference.	Environment/run record; no production test data.
Phase 2	Execute M1–M4 cases by role.	Authentication, course, approval and audience results.
Phase 3	Execute M5–M8 and available provider cases.	Exam, contest, media and regression results.
Phase 4	Log defects; fix separately; retest affected cases.	Defect list and retest evidence.
Phase 5	Review actual results and submit documentation.	Final plan and execution report; submission target 28 October 2026.

The schedule is a proposed sequence, not an execution claim. The 24-hour exam retake check and missing-provider cases may remain outstanding at today's submission; list their status honestly.

8. Tools & Dependencies

Tool	Version	Purpose
Next.js / React	14.2.35 / 18	Application runtime and interface under test.
Node.js / MongoDB	24.19.0 used / staging DB	Server runtime and isolated persistent test records.
Chrome + DevTools	Record installed version	Manual UI, responsive layout, Console and Network inspection.
Bunny Stream	Library 792504	Real upload/playback and protection checks; real video required.
SMS / Google OAuth	Configured service	OTP delivery and Google sign-in; Blocked without provider setup.
CSV / screenshots	Project results sheet	Actual outcomes, tester, defect IDs and redacted evidence.

9. Conclusion

This plan provides 69 manual cases for e-Sikho using the supplied report structure. It prioritizes role authorization, course entitlement, global hosting approval, audience eligibility and assessment integrity while covering the broader student/instructor workflows.

Manual execution remains Not Run until the team records actual outcomes. A completed report must identify every failed, blocked or unexecuted case and include evidence for fixes. Passing functional tests alone does not establish production readiness.

Real Bunny upload and protected playback need a real owned video. Production OTP/OAuth checks require working provider setup. Paid settlement is not available. Dependency advisories, the unsafe in-process code grader and limitations of public file URLs remain separate deployment concerns.

The companion results sheet maps TC-M identifiers to the original detailed manual cases. Keep passwords, OTPs, signed media URLs, API keys and database connection strings out of submission evidence.

Prepared By: _____ Date: _____

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